

Muhammad Imran

PERSONAL DATA

🏠 imran9pk.github.io | 🌐 github.com/imran9pk | 🎓 Google Scholar | 🔗 linkedin.com/in/imran9pk
📍 L'Aquila, Italy | 📞 +39 388 5682676 | ✉ imran9pk@gmail.com | muhammad.imran@graduate.univaq.it

EDUCATION

Ph.D. in Information and Communication Technologies Jun 2022 – Dec 2025

University of L'Aquila, Italy

Thesis: On Automating Performance Problems Detection in Software Systems through Empirical Analysis 📄

Advisors: Vittorio Cortellessa 🧑, Davide Di Ruscio 🧑, Luca Traini 🧑

Thesis Description

- Built empirical pipelines to automate performance problem analysis in software systems, covering test coverage, test prediction, bug detection, and energy estimation.
- Applied machine learning and large language models on static and dynamic code features, validated through large-scale Java experimentation.

Master of Science in Computer Science

Feb 2012 – Oct 2014

King Abdulaziz University, Saudi Arabia

Thesis: Agent Based Security Testing of Web Based Applications 📄

Advisor: Fathy Elbouraey Eassa 🧑

Bachelor of Science in Software Engineering

Sep 2006 – Jul 2010

International Islamic University Islamabad, Pakistan

INDUSTRY AND RESEARCH EXPERIENCE

Postdoc Researcher 🌐

Dec 2025 – Present

SPENCER Laboratory, University of L'Aquila, Italy

- Build and iterate empirical and ML pipelines for performance analysis on large Java systems, from dataset construction and feature extraction to model evaluation.
- Apply large language models to software engineering tasks, using controlled experiments and error analysis to improve measured outcomes.

Researcher (Scientific Collaborator) 🌐

Feb 2025 – Apr 2025

Software and Sustainability (S²) Group, Vrije Universiteit Amsterdam, Netherlands

- Designed and implemented an end to end Java method level energy estimation pipeline, including feature engineering, profiling, model training, and evaluation.
- Modeled energy behavior using static and dynamic features, iterating on feature sets and model design based on predictive performance.

Software Engineer 🔄

May 2011 – Nov 2011

Diyatech Pvt. Ltd., Islamabad, Pakistan

- Built and delivered modules for "Storage Edge", a SharePoint based product, implementing sprint scoped features under deadlines.
- Collaborated in an Agile team to integrate changes and deliver client aligned functionality.

Jr. Software Engineer 🔄

Sep 2010 – Apr 2011

360 Technologies Pvt. Ltd., Islamabad, Pakistan

- Implemented features for the Pakistan Navy Budget Management System, translating end user needs into working functionality.
- Supported requirements gathering, change requests, and iterative delivery with stakeholders.

PUBLICATIONS

Peer-Reviewed Publications

1. **Imran, M.**, Malavolta, I., & Stoico, V. *Static Metrics Are Insufficient: Predicting Java Method Energy Usage with Execution Time*. International Conference on the Quality of Information and Communications Technology (QUATIC 2026).  
2. **Imran, M.**, Cortellessa, V., Di Ruscio, D., Rubei, R., & Traini, L. *An Empirical Investigation on the use of Large Language Models for Performance Bug Detection*. IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2026).  
3. **Imran, M.**, Cortellessa, V., Di Ruscio, D., Rubei, R., & Traini, L. *Is Code Coverage of Performance Tests Related to Source Code Features? An Empirical Study on Open-Source Java Systems*. *Empirical Software Engineering (EMSE)*, Volume 30, article number 157, (2025).  
4. **Imran, M.**, Cortellessa, V., Di Ruscio, D., Rubei, R., & Traini, L. *An empirical study on code coverage of performance testing*. International Conference on Evaluation and Assessment in Software Engineering (EASE 2024).  
5. **Imran, M.**, Jambi, K., & Eassa, F. *Dynamic Analysis for Security Testing of Web-Based Applications Using Agent Technology*. Fourth Intl. Conf. Advances in Computing, Communication and Information Technology (CCIT 2016). 
6. Khan, I. H., Siddiqui, M. A., Jambi, K. M., **Imran, M.**, & Bagais, A. A. *Query optimization in Arabic plagiarism detection: An empirical study*. International Journal of Intelligent Systems and Applications, 7(1), 73–79 (2015). 
7. **Imran, M.**, & Alhalabi, W. *Improving the operating system resource management by processes simulation*. International Journal of Scientific & Engineering Research, 5(2), 1404–1408 (2014). 

TEACHING EXPERIENCE

Lecturer

Feb 2019 – May 2022

International Islamic University, Islamabad, Pakistan

- Taught undergraduate CS and SE courses including Software Engineering, Requirements Engineering, Object-Oriented Analysis and Design, and Software Testing.
- Supervised final year projects and contributed to departmental accreditation committees.

Lecturer

Jul 2015 – Jan 2019

Iqra University, Islamabad Campus, Pakistan

- Delivered undergraduate CS courses and supervised final year projects with academic and technical mentorship.
- Coordinated undergraduate projects, managing workflows and faculty-student alignment.

ACADEMIC SERVICE

Program Committees and Reviewing

- **ISSTA 2026** (ACM SIGSOFT International Symposium on Software Testing and Analysis), Artifact Evaluation Committee Member, California, United States. 
- **ASE 2026** (IEEE/ACM International Conference on Automated Software Engineering), Artifact Evaluation Committee Member. 
- **MSR 2026** (International Conference on Mining Software Repositories), Junior Program Committee Member, Technical Papers track, Rio de Janeiro, Brazil. 
- **ECSA 2026** (European Conference on Software Architecture), Open Science Committee Member, Bolzano, Italy. 
- **AISA 2025** (International Workshop on AI-Assisted Software Architecting), Program Committee Member, co-located with ECSA 2025, Limassol, Cyprus. 

HONOURS AND AWARDS

Ph.D. Scholarship

2022 – 2025

University of L'Aquila, Italy — Awarded for doctoral studies in Computer Science

International Scholarship (MSc)

2012 – 2014

King Abdulaziz University, Saudi Arabia — Awarded for academic excellence

Merit Scholarship (BSc)

2007 – 2010

International Islamic University, Pakistan — Awarded for high academic performance

REFERENCES

1. **Prof. Vittorio Cortellessa** (PhD Advisor) 
Professor, University of L'Aquila, Italy
vittorio.cortellessa@univaq.it
2. **Prof. Fathy Elboursaey Eassa** (MSc Advisor) 
Professor, King Abdulaziz University, Jeddah, Saudi Arabia
fathy55@yahoo.com
3. **Assoc. Prof. Ivano Malavolta** (Scientific Collaborator) 
Associate Professor, Vrije Universiteit Amsterdam, Netherlands
i.malavolta@vu.nl

Last Update: July 22, 2026